

S. VANDHANA

+91-8925076444 • vandhanasatish16@gmail.com • LeetCode • LinkedIn.in/Vandhana • GitHub/Vandhana

PROFESSIONAL SUMMARY

Computer Science and Engineering graduate with hands on experience in Python, SQL, FastAPI, REST APIs, Machine Learning, and Deep Learning. Built a dual model CNN system for medical image classification with FastAPI, SQLite, and JWT authentication, along with data analytics and full stack projects. Completed an internship in automation and Python based development, with a strong foundation in Operating Systems, Computer Networks, and Linux.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL
- **Web and Backend:** FastAPI, REST APIs, JWT Authentication, HTML, CSS
- **Machine Learning and AI:** Machine Learning, Deep Learning, TensorFlow, Scikit learn
- **Data and Databases:** Pandas, NumPy, Power BI, Microsoft Excel
- **Tools and Platforms:** Git, GitHub, Linux, Streamlit
- **Core Computer Science:** Operating Systems, Computer Networks, DBMS, Object Oriented Programming (OOP), Data Structures and Algorithms

PROJECTS

1. BrainNet AI - Brain Tumor MRI Classification System | [GitHub](#)

Tech: Python, Keras, FastAPI, SQLite, JWT

- Built a dual model deep learning system (EfficientNetB0, ResNet50V2) achieving 94.6% accuracy on a 4 class brain tumor MRI dataset of 5,600 training and 1,600 testing images
- Designed a novel Spatial Consensus Confidence (SCC) trust metric using IoU of GradCAM++ heatmaps, improving prediction reliability, high confidence cases reached 92.7% accuracy vs. 77.2% for low confidence cases (a 15.5 percentage point trust gap)
- Developed a FastAPI backend with JWT authentication, SQLite based scan history, and downloadable PDF clinical reports

2. Customer Shopping Behavior Analysis - End to End Analytics Pipeline | [GitHub](#)

Tech: Python, SQL, Power BI

- Built a 3 stage analytics pipeline (Python for cleaning/EDA, SQL for segmentation queries, Power BI for dashboarding) replicating an enterprise analytics workflow
- Performed data cleaning and exploratory analysis on customer shopping data to identify purchase patterns and retention factors
- Designed an interactive Power BI dashboard translating SQL derived customer segments into visual business insights for decision making

3. Restaurant Ordering Web Application | [GitHub](#)

Tech: HTML, CSS, JavaScript

- Built a responsive restaurant ordering web application enabling users to browse menu items by category, add items to a cart, and complete a simulated checkout flow
- Implemented dynamic cart functionality using vanilla JavaScript, enabling real time addition, removal, and quantity updates without page reloads
- Designed a mobile responsive user interface ensuring a consistent experience across desktop and mobile devices.

EDUCATION

Btech CSE with specialization in Bioinformatics

2026

Vellore Institute of Technology | CGPA: 8.17/ 10

Senior Secondary

2022

SBOA School and Junior College | 80%

Secondary

2020

SBOA School and Junior College | 81.8%

CERTIFICATES

DeepLearning.AI, [Generative AI](#)

Anthropic, [AI Fluency: Frameworks and Foundations](#)

IBM, [CyberSecurity Analyst](#)